Advanced Biotechnology

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CEBT II - Michaelmas 2025



UNIVERSITY OF CAMBRIDGE

Lecture 11

Bioprocessing

Cell disruption methods (part II)

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Primary isolation methods (part I)



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Lecture outline

1) Cell disruption methods: Physical (non-mechanical), Chemical and Biological methods
2) Overview of primary isolation
3) Methods for primary isolation
   I. Solvent extraction
   II. Aqueous two-phase extraction
   III. Precipitation (salting out, isoelectric precipitation, polymer precipitation, solvent precipitation)

Chapter 11 of Bioprocess Engineering Principles (11.5, 11.7, 11.8)

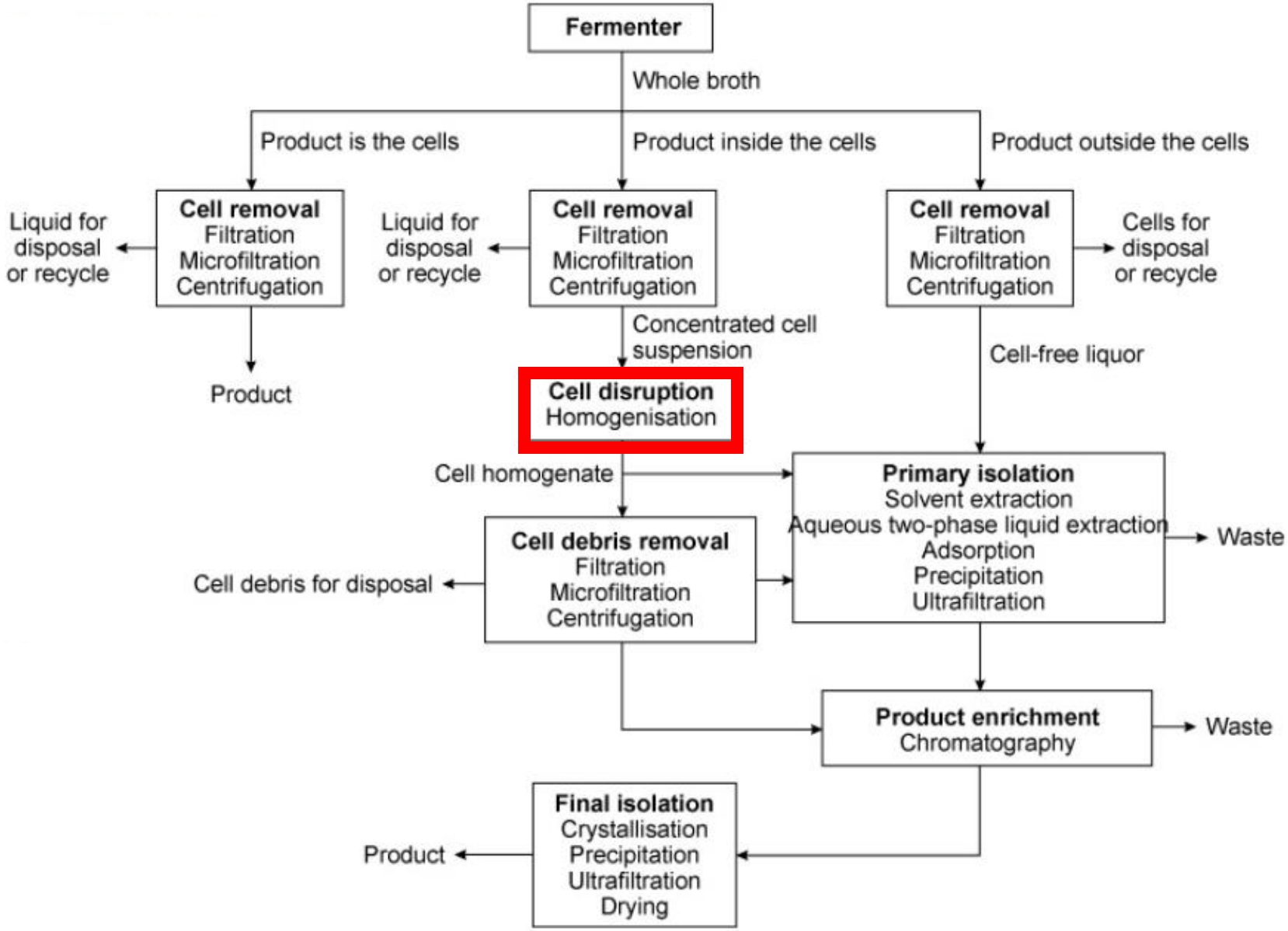
Learning outcomes

1) Discuss principles, advantages and disadvantages of various non-mechanical cell disruption methods
2) Discuss purpose of primary isolation in the context of downstream processing
3) Discuss principles, pros, cons and applicability of primary isolation methods: solvent extraction, two-phase aqueous extraction and precipitation
4) Discuss various strategies to induce protein precipitation
5) Mathematically describe efficiency of some primary isolation methods (solvent and aqueous two-phase extraction, precipitation via salting out)

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1) **Cell disruption methods. Non-mechanical physical, Chemical and Biological methods**
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Downstream process logic



Cell-disruption strategies

- **Mechanical methods (previous lecture)**
  - Bead mill
  - Ultrasonication
  - French press
  - High pressure homogenizer
- **Non-mechanical physical methods (this lecture)**
  - Thermolysis
  - Decompression
  - Osmotic shock
- **Chemical methods (this lecture)**
  - Detergents
  - Solvents
- **Biological methods (this lecture)**
  - Enzymatic digestion

Non-mechanical physical methods

- Include **thermal**, **decompression** and **osmotic** shock
- Costly and difficult to scale up, therefore used mostly at lab rather than production scale

Non-mechanical physical methods: Thermolysis

- Cells can be disrupted by **heating**. *E.g.* cytoplasmic proteins can be extracted form *E. coli* if kept at 90°C for 10 minutes
- Cells can be disrupted by **freeze-thaw cycles**, due to formation of ice crystals that disrupt cell membranes/walls
- Both these techniques are **energy intensive** and therefore not very suitable for production scale processes

Non-mechanical physical methods: Decompression

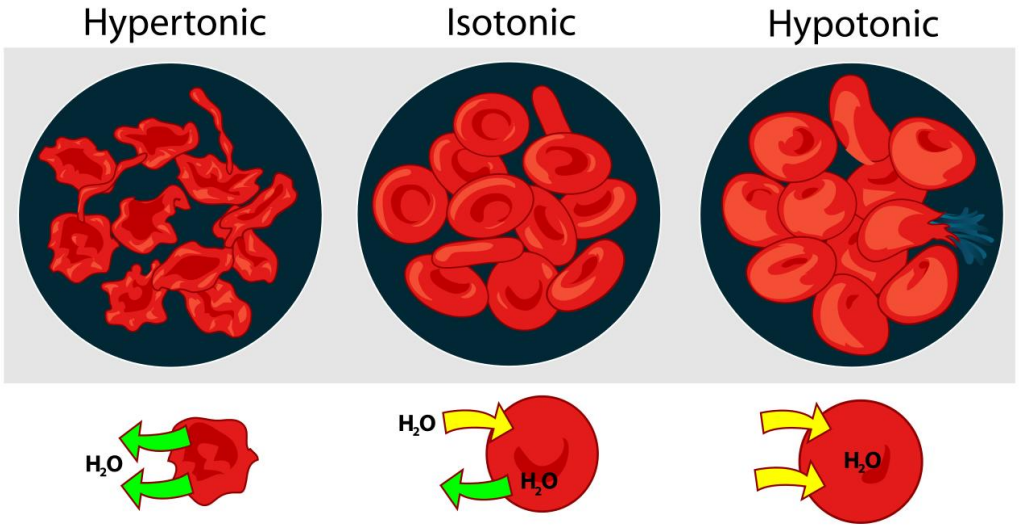
- Cells are exposed to **high-pressure gas** that solubilizes in the medium and distributes itself also within the cells
- Upon **rapid decompression**, the gas within the cells expands, causing disruption



Decompression chamber (Anton-Parr)

Non-mechanical physical methods: Osmotic shock

- Cells exposed to **hypertonic** or **hypotonic** conditions.
- **Hypertonic** conditions are typically obtained by adding salts (~1M concentration) and cause the cells to shrink rapidly as water leaves the cytoplasm. The process may disrupt the cells
- **Hypotonic** conditions are obtained by diluting in pure water, causing water to flow into the cytoplasm. Cells can burst in the process
- Not sufficiently efficient for production-scale. Added salt also needs to be removed, which adds to cost

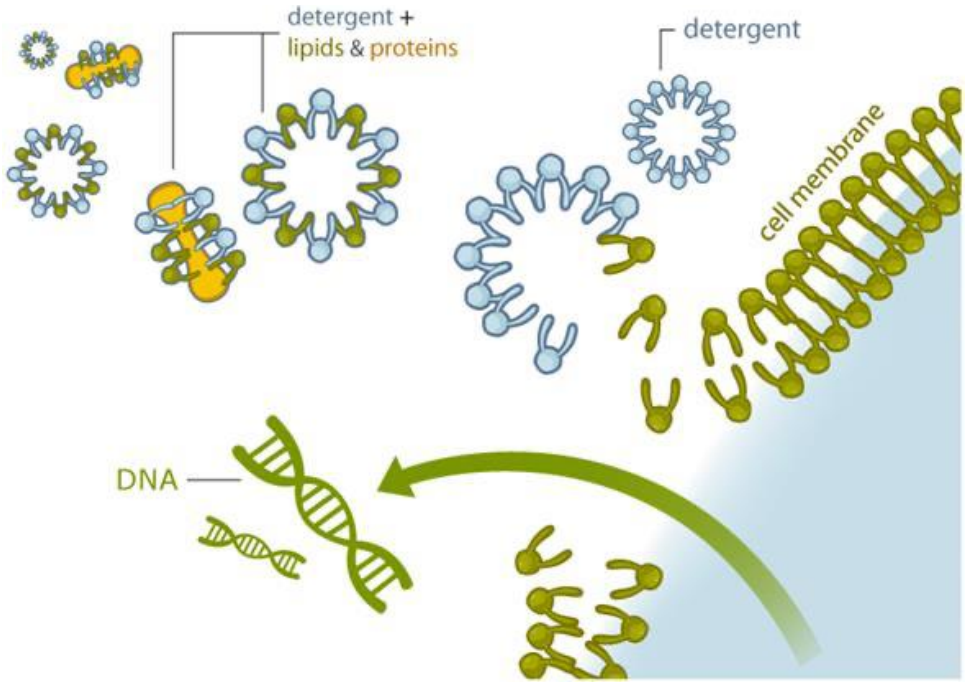


Non-physical (chemical and biological) methods

- Non-physical methods disrupt cell walls and membranes through the addition of chemical substances (**solvents**, **detergents**) or biological agents(**enzymes**)
- Multiple substances can be combined to achieve the desired efficiency
- Drawbacks compared to physical methods is the cost of the substances used, and that these may need to be removed

Chemical methods: Detergents

- Detergents, or **surfactants**, can solubilize the phospholipid cell membranes through micellization and by solubilizing membrane proteins
- Typical detergents used are sodium dodecyl sulfate (**SDS**) and **Triton X100**
- Detergents may need to be removed for later processing stages, adding to cost
- Commonly used at lab-scale for RNA or DNA extraction



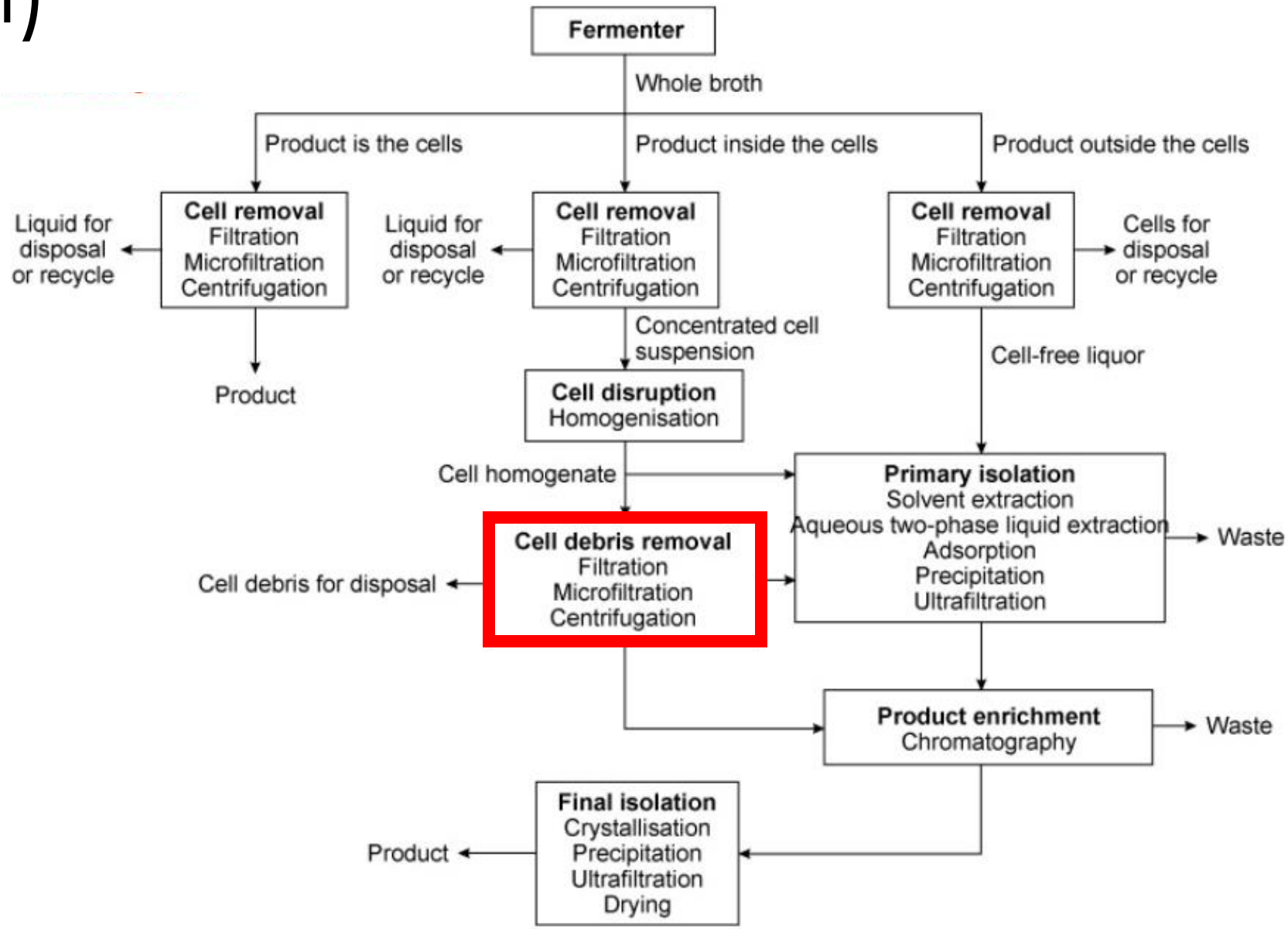
Chemical methods: Solvents

- Solvents such as **alcohols**, **dimethyl sulfoxide**, **methyl ethyl ketone** or **toluene** can solubilize the lipid components in the cell membrane, causing their disruption
- Like detergents, also solvents may need to be removed for later processing stages
- Solvents **may cause denaturation** of target proteins, which may be undesirable
- Use of solvents may be combined with alkaline compounds used to hydrolyze the cell wall

Biological methods: Enzymes

- Utilizes enzymes to digest cell walls
- Enzymes include **lysozyme**, **cellulase, protease**, **chitinase**, **pectinase** (and more)
- It is specific: different enzymes can be used to digest different parts of the cell wall. For instance lysozyme is used to hydrolyze the peptidoglycan layer in bacteria, chitinase for yeast cell wall and pectinase for plant cell wall
- Can be combined with detergents. *E.g.* for gram – bacteria lysozyme treatment follows detergent treatment used to remove the outer membrane and expose the peptidoglycan layer
- Enzymes are costly, limiting application in production-scale processes

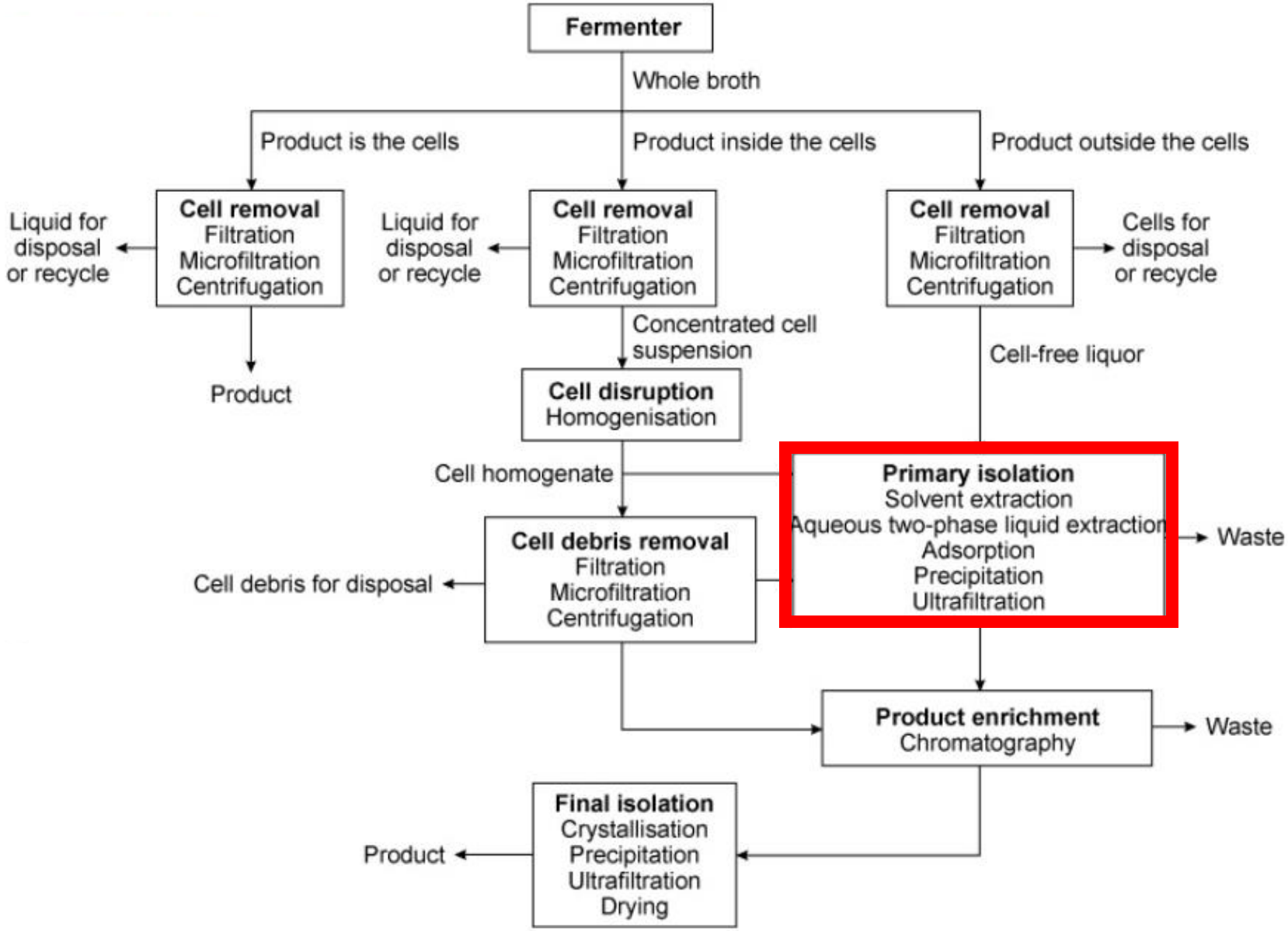
Cell-debris removal (centrifugation, filtration)



Lecture outline

1) Cell disruption methods. Non-mechanical physical, Chemical and Biological methods
2) **Overview of primary isolation**
3) Methods for primary isolation
   I. Solvent extraction
   II. Aqueous two-phase extraction
   III. Precipitation (salting out, isoelectric precipitation, polymer precipitation, solvent precipitation)

Downstream process logic



Primary Isolation

- We are left with feed that is either a **liquor from which the micro-organisms have been removed OR a cell homogenate**

- The target products of interest (*e.g.* proteins, peptides, small molecules such as antibiotics…) are in solution along with many non-desirable components

- **The aim of primary isolation stage is to remove components with substantially different properties to the product of interest**

- Primary isolation processes are usually not very selective and **depend on the chemical and physical properties of the product**

Lecture outline

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   I. **Solvent extraction**
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Primary Isolation methods

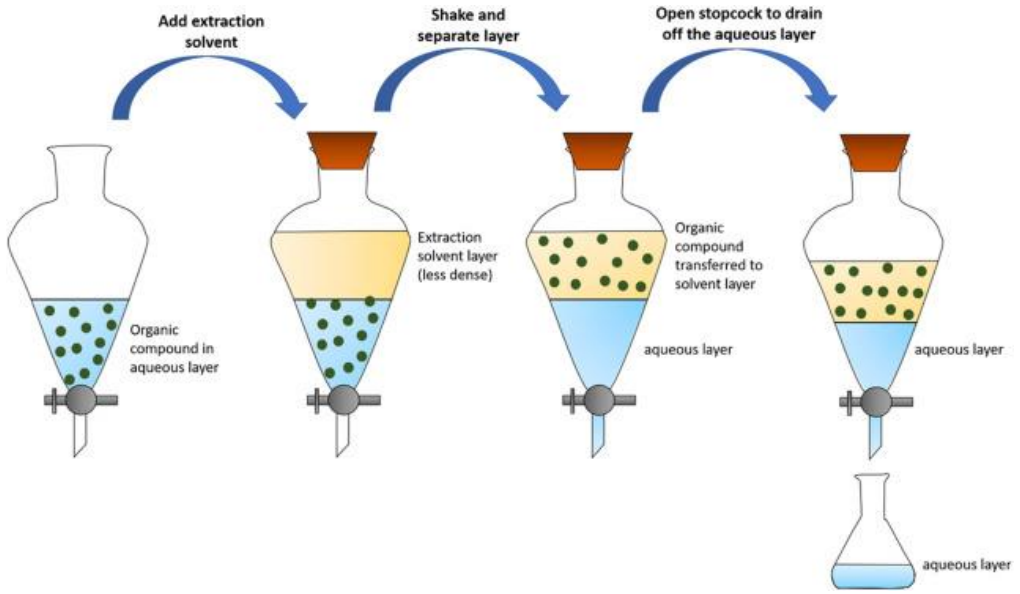
Typical operations include (but are not limited to)

I. Solvent extraction (this lecture)
II. Aqueous two-phase extraction (this lecture)
III. Precipitation (this lecture)
IV. Adsorption (next lecture)

In this lecture and part of the next we are going to give an overview of the principles and applicability of these methods

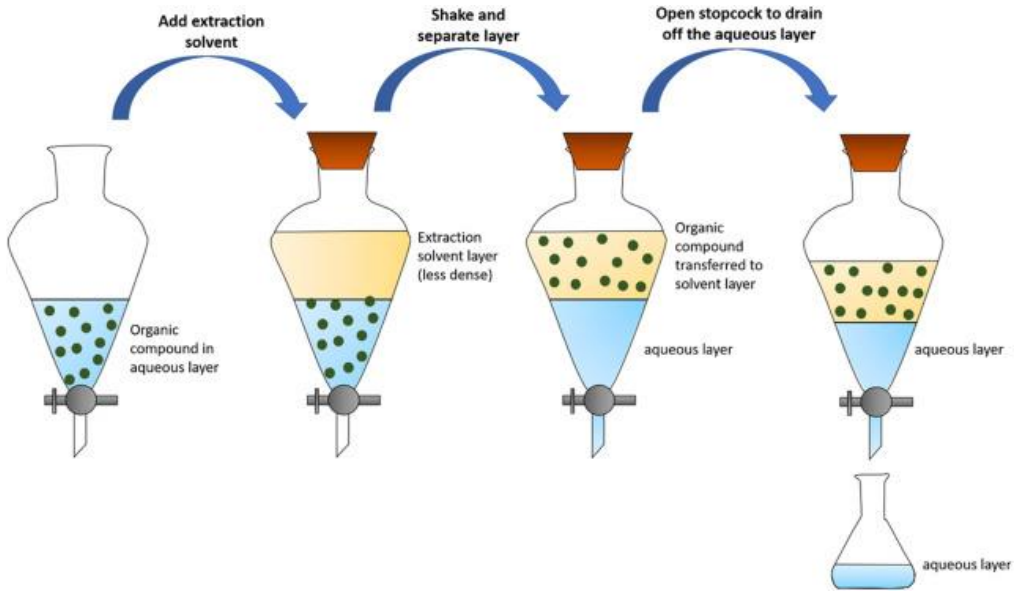
I. Solvent extraction

- The aqueous feed contains a target organic compound A
- It is mixed with an organic solvent, immiscible with the aqueous feed, within which A preferentially partitions
- The system is agitated to allow transfer of A to the solvent and left to settle
- Solvent and aqueous phase are separated
- A is recovered from the solvent, *e.g.* through evaporation



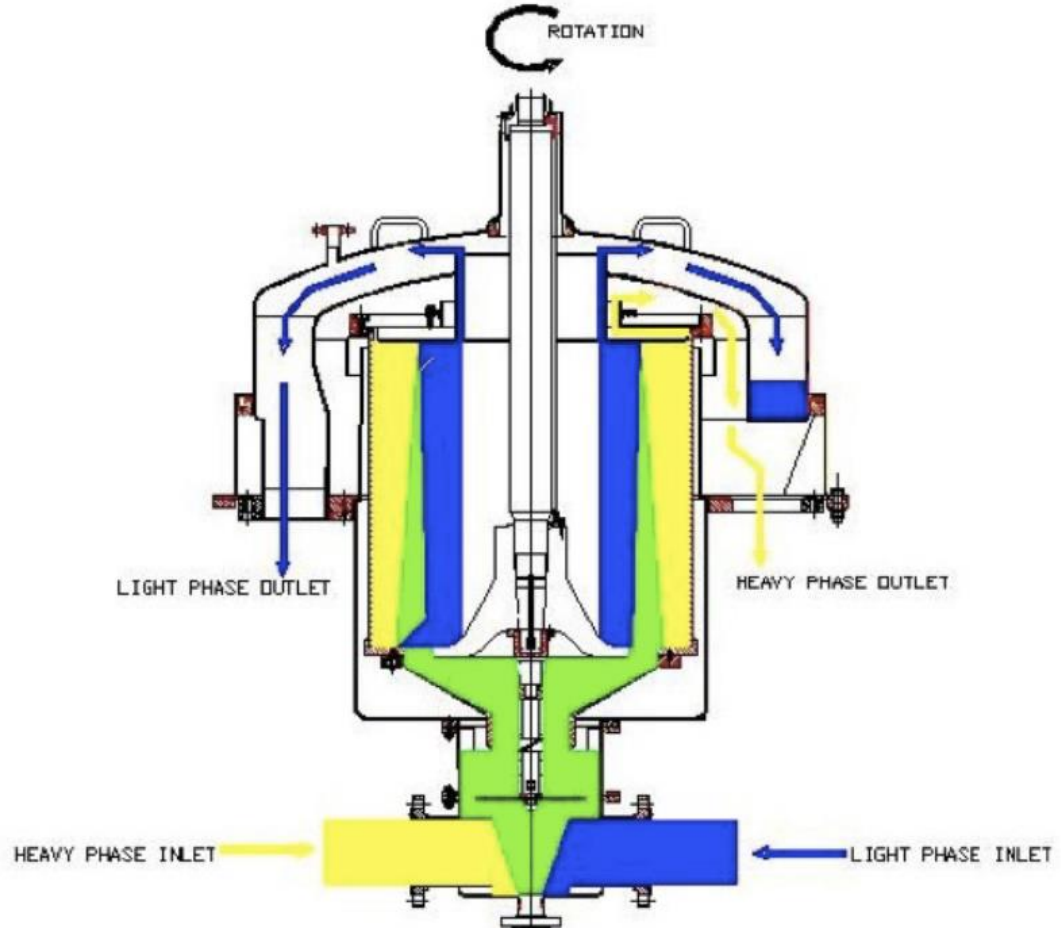
I. Solvent extraction

- Solvent is chosen so that A partitions strongly within it, but undesirable compounds in the feed (*e.g.* cell debris) do not.
- Often used for extracting biopharmaceuticals
- *E.g.* penicillin can be extracted from aqueous broths using solvents such as butyl or amyl acetate.



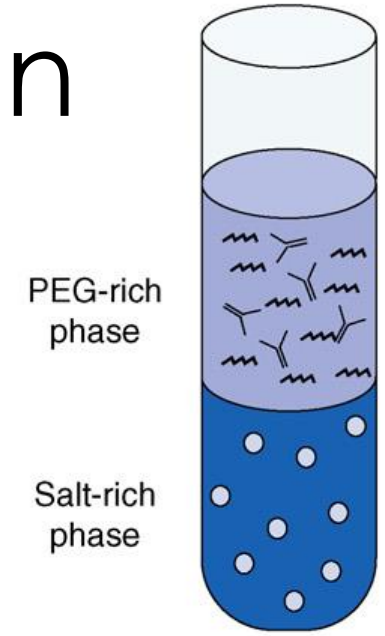
I. Solvent extraction

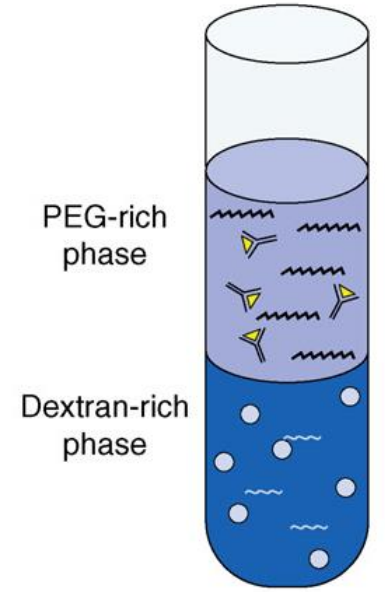
- In a plant-scale process, solvent extraction is often carried out in continuous mode rather than batch mode
- A **countercurrent centrifuge** can be used for this. The heavy phase (typically the feed) is mixed with the light phase (typically solvent) by agitators at the bottom, enabling transfer of the target compound from one phase to the other. A centrifuge then separates and recovers the two phases



II. Aqueous two-phase extraction

- Similar principle to solvent extraction, but here we exploit liquid-liquid phase separation between **two aqueous phases** rather than organic solvents
- Phase separation can be induced by adding solutes such as **polymers of different chemistry**, and salts
- Solutions contacting **poly-ethylene glycol (PEG) and salts** will separate in salt-rich and PEG rich phases (top)
- Solutions of **PEG+Dextran** will separate in PEG-rich and Dextran-rich phases (bottom)

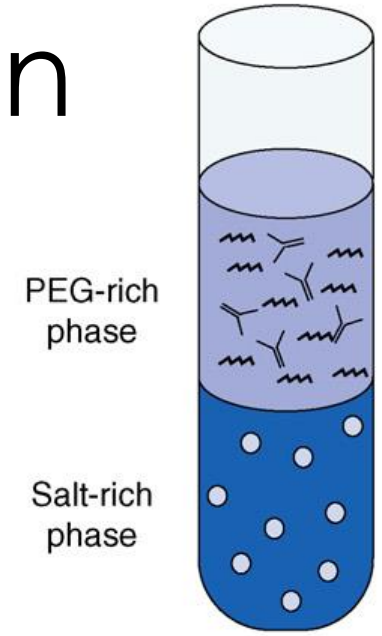


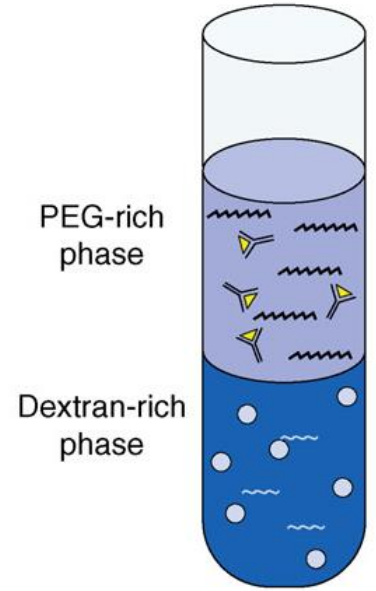


Azevedo et al, Trends in Biotechnology 27, 4 2009

II. Aqueous two-phase extraction

- Often used in pharma, *e.g.* for separation of monoclonal antibodies, growth factors and hormones
- Target compounds will **partition** in one or the other phases **depending on their physical and chemical characteristics**
- Degree of partitioning can be fine tuned by changing system composition (adding salts, changing polymer MW)
- Easily scalable





Azevedo et al, Trends in Biotechnology 27, 4 2009

Aqueous and solvent two-phase separation theory

- The partition between the upper and lower phase can be expressed in terms of a **partition coefficient**, K

$$K = \frac{C_{A,u}}{C_{A,l}}$$

where $C_{A,u}$ and $C_{A,l}$ are the equilibrium concentrations of component A in the upper and lower phase, respectively

- Applies to both solvent-aqueous aqueous-aqueous systems

Aqueous and solvent two-phase separation theory

- A **very large or very small K is desirable for good separation**, but we can also tune the relative volume of the two phases
- The **yield** (for recovery of A in the upper phase), or the **total fraction of recovered compound**, can be defined as

$$Y_{A,u} = \frac{V_u C_{A,u}}{V_0 C_{A,0}} = \frac{V_u C_{A,u}}{V_u C_{A,u} + V_l C_{A,l}} = \frac{\frac{V_u}{V_l}}{\frac{V_u}{V_l} + \frac{1}{K}}$$

where V_u is the volume of the upper phase, $V_0 = V_u + V_l$ is the total volume of the system, $C_{A,0}$ the average concentration of A in the system, and $C_{A,u}$ the concentration of A in the upper phase

- We can therefore increase the yield by increasing the volume of the upper phase.
- Similar yield can be defined for the lower phase

Example: enzyme recovery using aqueous two-phase extraction

Aqueous two-phase extraction is used to recover α-amylase from solution. A PEG-dextran mixture is added and the solution separates into two phases: PEG-rich upper phase and Dextran-rich lower phase. The partition coefficient is $K = \frac{C_{A,u}}{C_{A,l}} = 4.2$. Calculate the maximum possible enzyme recovery when:

a) The volume ratio of upper to lower phase is 5.0

b) The volume ratio of upper to lower phase is 0.5

Solution

We use the expression for $Y_{A,u}$ in terms of $\frac{V_u}{V_l}$ and K

$$Y_{A,u} = \frac{\frac{V_u}{V_l}}{\frac{V_u}{V_l} + \frac{1}{K}}$$

Substituting the numbers, we obtain for

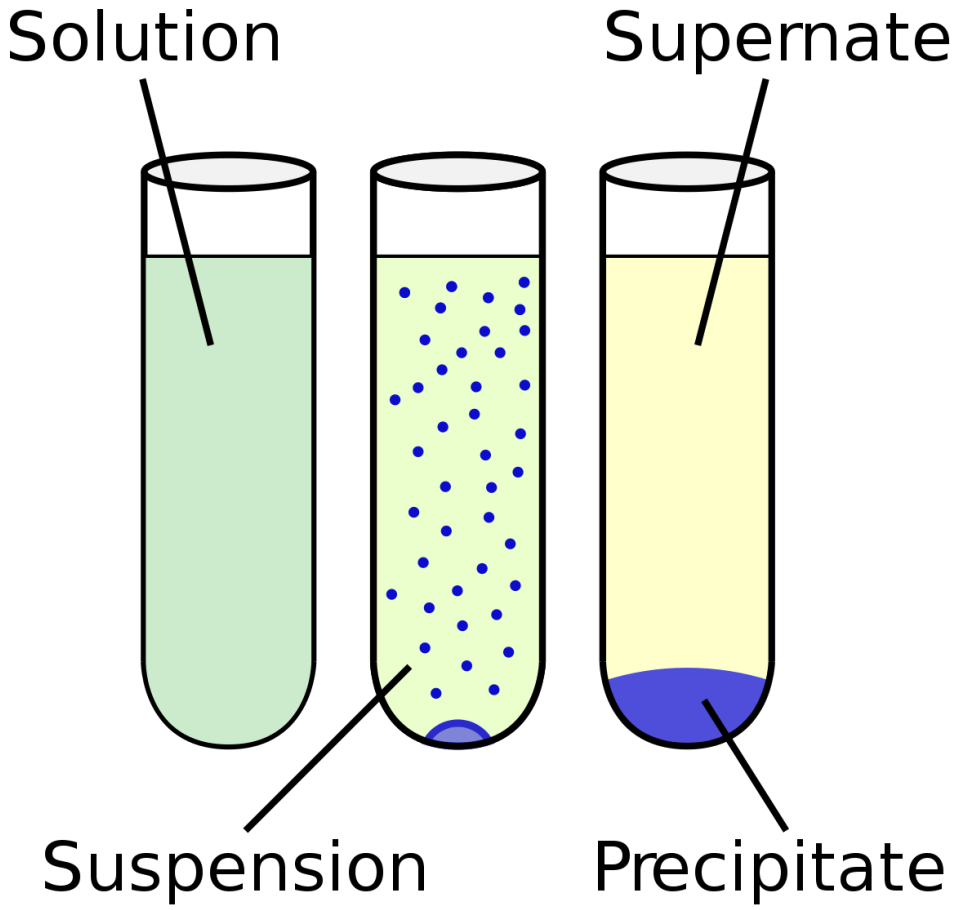
a) $Y_{A,u} = 0.95$ $(\frac{V_u}{V_l} = 5)$

b) $Y_{A,u} = 0.68$ $(\frac{V_u}{V_l} = 0.5)$

Clearly, the yield is much higher for case a, where the volume of the upper phase is much in excess compared to the lower phase

III. Precipitation

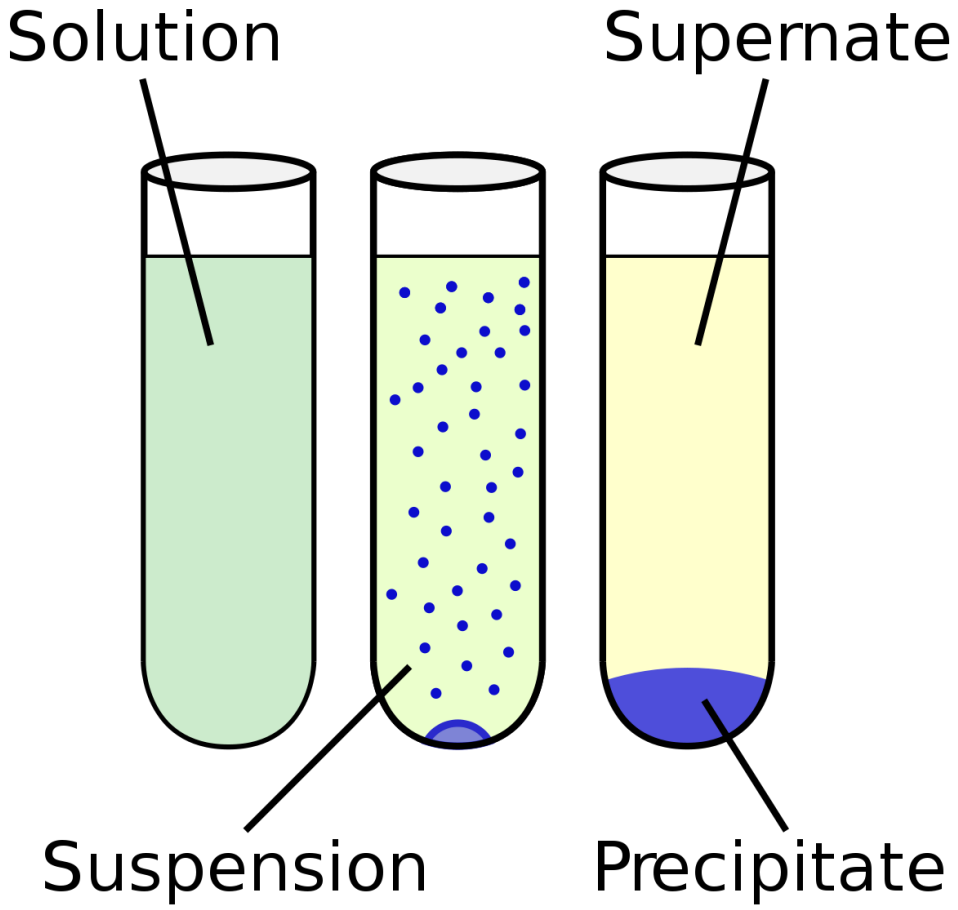
- We induce **aggregation of the target compound**, which then precipitates and can be easily extracted using *e.g.* further filtration or centrifugation.
- **Often used for proteins**
- Precipitation can remove large amounts of water
- However, only partial purification is achieved since molecules of similar characteristics will precipitate together



Wikimedia commons

III. Precipitation

- Precipitation is induced by a **decrease in the solubility** of the compound upon addition of **salts**, **solvents** or **polymers**, or in response to changes in **pH**
- These additives or changes in surface chemistry of the compounds **make interactions with water less favorable** causing aggregation



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III. Precipitation

In bioprocessing, there are four main methods used to achieve precipitation, primarily to recover **proteins**. These are:

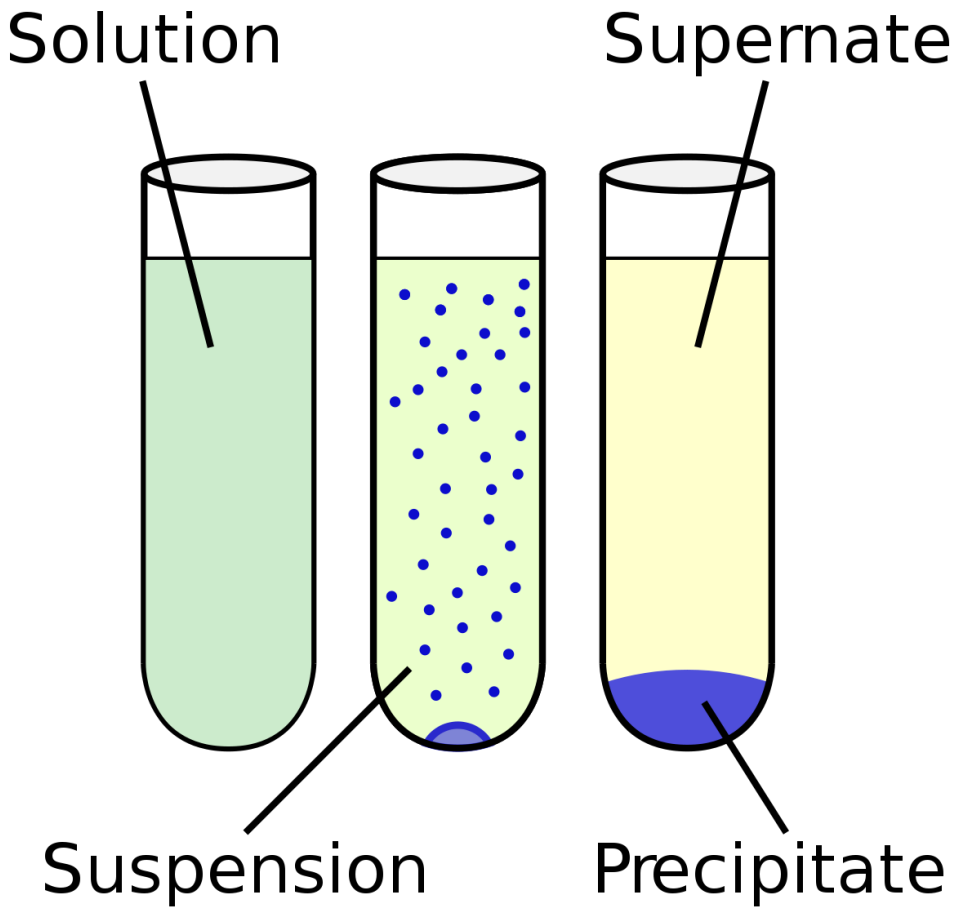
a) Salting out

b) Isoelectric precipitation

c) Polymer precipitation

d) Organic solvent precipitation

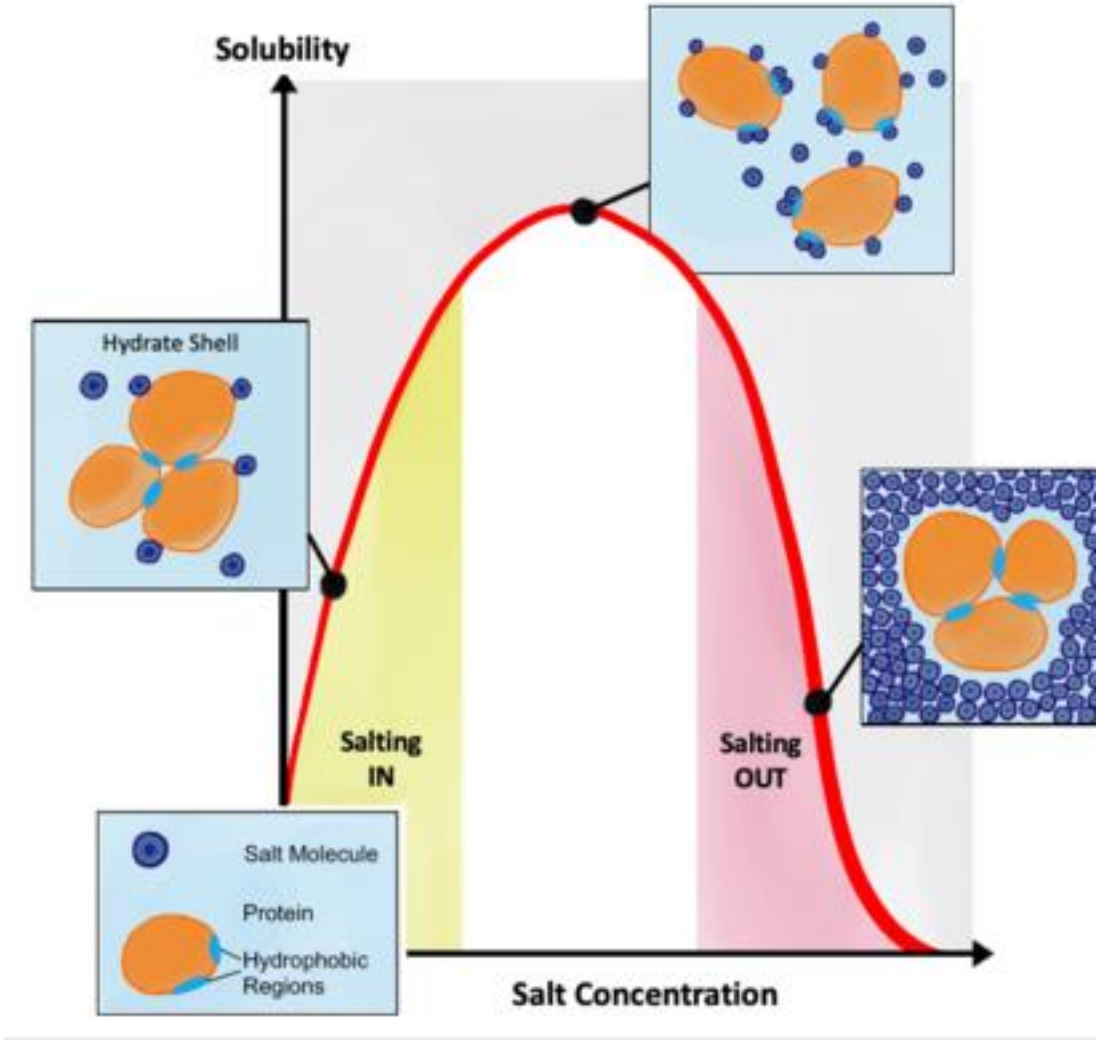
Let's discuss them one by one



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III. Precipitation: a) Salting out

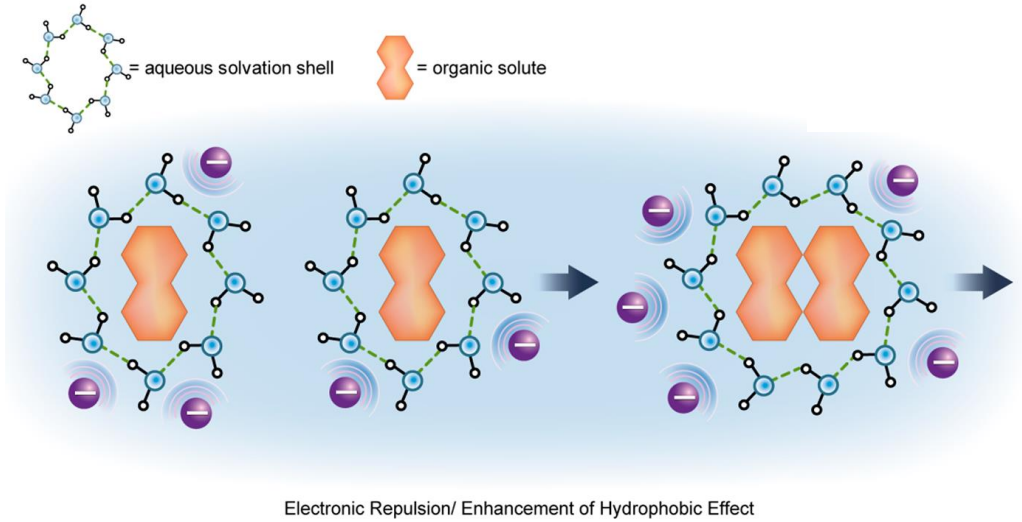
- **Solubility of molecules with hydrophobic regions (*e.g.* proteins) depends on salt concentration**.
- Solubility is low at low salt concentration, peaks at intermediate concentration, and then drops again at very high concentration.
- The process of resuspending molecules by adding salt is referred to as **salting-in**, while that causing precipitation is called **salting-out**, and is relevant for protein precipitation in bioprocessing
- Different proteins will salt out at different salt concentrations depending on their hydrophobicity.



Wikimedia commons

III. Precipitation: a) salting out

- Salting out happens because ions, and particularly **anions**, interfere with solvation by water
- Water molecules form a **solvation shell** around proteins, which **prevents aggregation**
- At higher salt concentration, anions will bind more often to **water**, making it **less available to form the solvation** shell and thus prompting aggregation
- Solvation is also important for the secondary and tertiary structure of proteins. **High salt may also cause protein unfolding**



Org. Process Res. Dev. 2017, 21, 1355−1370

III. Precipitation: a) salting out

- Different anions have different effectiveness for inducing salting out.
- They are normally ranked as:

$$F^- \approx SO_4^- > HPO_4^- > C_2H_3O_2^- > Cl^- > Br^- > NO_3^- > ClO_3^- > I^- > ClO_4^- > SCN^-$$

- This was discovered by Franz Hofmeister, and it is known as the **Hofmeister series**.
- The origin of these differences is still matter of research!



Franz Hofmeister - Wikimedia commons

III. Precipitation: a) salting out

The **Cohn equation** describes the solubility S of a protein in the salting-out regime

$$\ln S = \beta_S - K_S C_S$$

where β_S and K_S are constants that depend on the protein and salt used, and C_S is the salt concentration. β_S is also a function of temperature and pH

Example: antibody recovery by salting-out

Pure IgA monoclonal antibody in has solubility of $770\ \mu g\ ml^{-1}$ in $1.4\ M\ (NH_4)_2SO_4$ and $29\ \mu g\ ml^{-1}$ in $3.0\ M\ (NH_4)_2SO_4$.

A solution containing $180\ g$ of pure antibody is treated for protein recovery using $2.5\ M\ (NH_4)_2SO_4$ in a total volume of $1000\ l$.

What percentage of the antibody would precipitate?

Solution

To answer, we need to derive the solubility of the protein at the target salt concentration, using the Chon equation

$$\ln S = \beta_S - K_S C_S$$

To do so, we need to extract β_S and K_S

We are given the solubility (S_1 and S_2) for two salt concentrations ($C_{S,1}$ and $C_{S,2}$). We note that solubility decreases with increasing salt, so we are in the **salting-out regime**

For both the Cohn equation applies with the same β_S and K_S

$$\ln S_1 = \beta_S - K_S C_{S,1}$$
$$\ln S_2 = \beta_S - K_S C_{S,2}$$

We have two equations in two unknowns, so we can compute β_S and K_S

Solution

Subtratcing the the two equations to obtain

$$\ln\frac{S_1}{S_2} = K_s C_{S,2} - K_S C_{S,1}$$

From which we can extract:

$$K_S = \frac{\ln\frac{S_1}{S_2}}{C_{S,2} - C_{S,1}} = \frac{\ln\frac{770}{29}}{3\ -1.4} = 2.05\ \mathrm{M}^{-1}$$

Then solve for β_S,

$$\beta_S = \ln S_1 + K_S C_{s,1} = \ln 770 + 2.05 \times 1.4 = 9.52$$

Solution

Then, evaluate solubility of the protein when $C_S = 2.5\ \mathrm{M}$,

$$\ln S = 9.52 - 2.5(2.5)$$

$$S = 81.5\ \mu\mathrm{g\ ml^{-1}}$$

We need to compare the solubility with the concentration of our sample. We have $180\ \mathrm{g}$ in $1000\ \mathrm{l}$, equivalent to $180\ \mu\mathrm{g\ ml^{-1}}$

Of this, only $180\ \mu\mathrm{g\ ml^{-1}} - S = 98.5\ \mu\mathrm{g\ ml^{-1}}$ would precipitate corresponding to $98.5\ \mathrm{g}$ or 55%. This is very poor separation!

Solution

55% is very poor separation!

What salt concentration would one need to recover 99% of the antibody? How does this compare with the solubility of the salt?

III. Precipitation: b) isoelectric

- Proteins most often carry net **electric charge** or are zwitterionic (net charge is zero, but parts of the protein are positively or negatively charged)
- **Charge** is important to **stabilize proteins** against aggregation, due to Coulomb repulsion that counteracts hydrophobic interactions
- **Charge is pH dependent** due to (de)protonation of charged amino acids. At low pH it is more likely that a protein be positively charged. At high pH it is more likely negatively charged

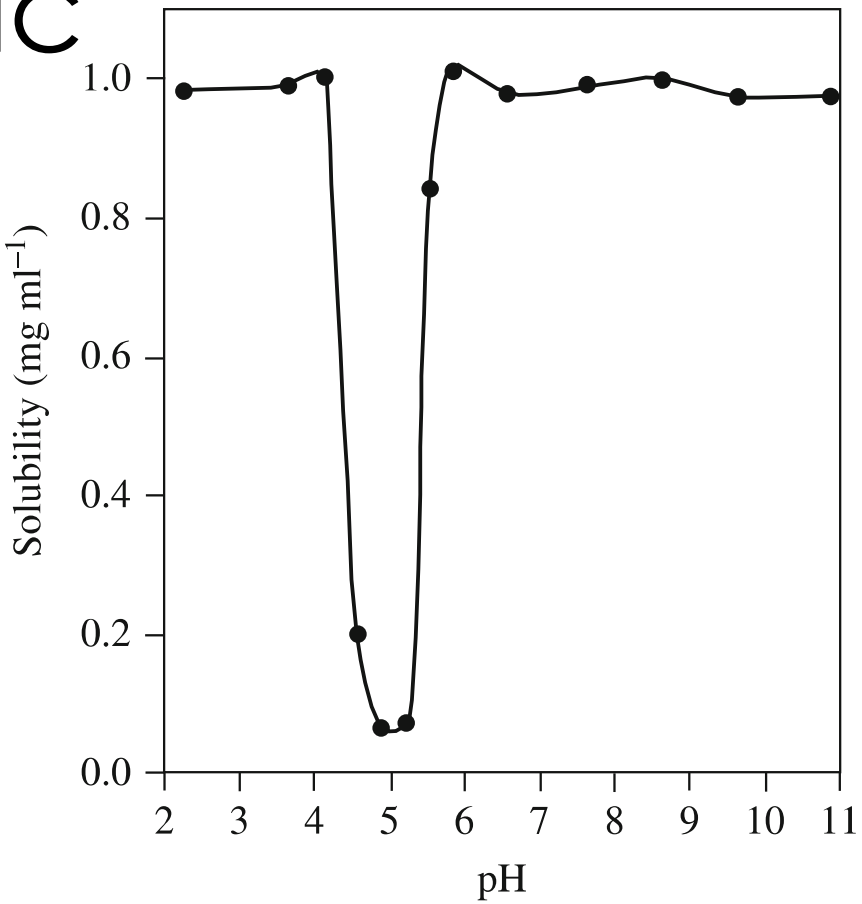
III. Precipitation: b) isoelectric

- The Isoelectric point, pI, is the pH at which a given protein is neutral
- Different amino acids have different pI (right)
- The pI of a protein depends on its composition, but it is usually between 4 and 6

no.	amino acid	pI
1	alanine	6.11
2	arginine	10.76
3	asparagine	5.43
4	aspartic acid	2.98
5	cysteine	5.15
6	glutamic acid	3.08
7	glutamine	5.65
8	glyeine	6.06
9	histidine	7.64
10	isoleucine	6.04
11	leucine	6.04
12	lysine	9.47
13	methionine	5.71
14	phenylalanine	5.76
15	proline	6.3
16	serine	5.7
17	threonine	5.6
18	tryptophan	5.88
19	tyrosine	5.63
20	valine	6.02

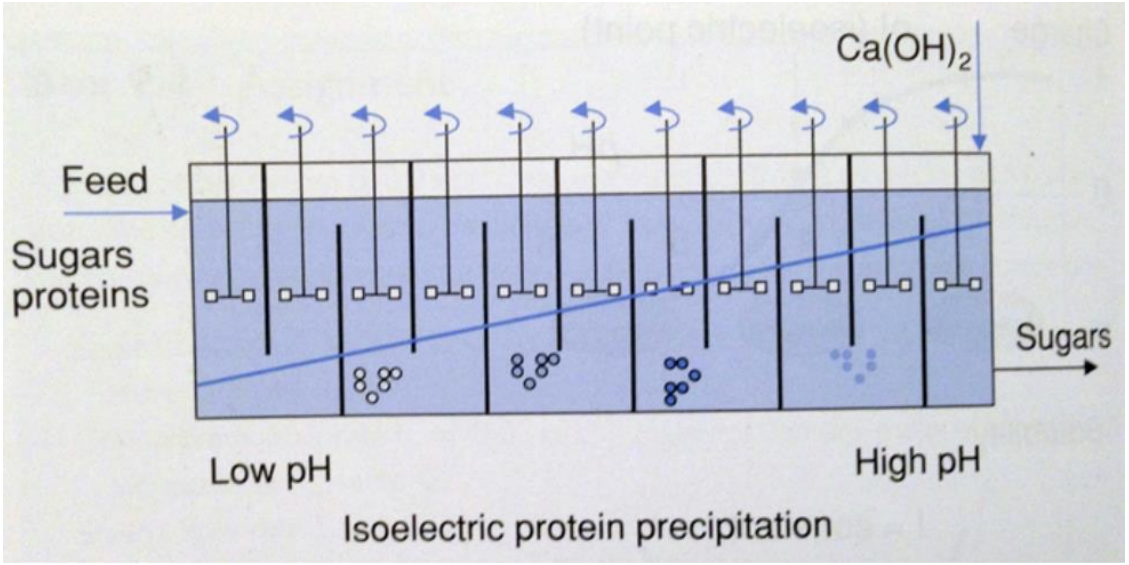
III. Precipitation: b) isoelectric

- Solubility of proteins as a function of pH therefore has a minimum as shown on the right
- Isoelectric precipitation works by changing (typically reducing) the pH of the feed until the protein of interest reaches minimum solubility and precipitates
- Advantage over salting out: de-salting step is not required
- Drawback: similar to salting out, reaching the isoelectric point **may lead to protein denaturation**



Solubility of soybean vicilin (7S) protein as a function of pH. The protein isoelectric point is close to pH 5.0.
J. Agric. Food Chem. 50, 49534958.

Example of isoelectric precipitation: removing proteins in sugar processing



- Washing, grinding and filtering **sugar beets** leaves a feed rich in sugar and various proteins
- We wish to precipitate the proteins and keep the sugar
- A gradient of pH is established in a horizontal tank with stirrers
- In each compartment a different protein pI equal to the pH at that point precipitates
- A protein-free sugar solution is finally produced
- Process used at **British Sugar plant in Bury St Edmunds**

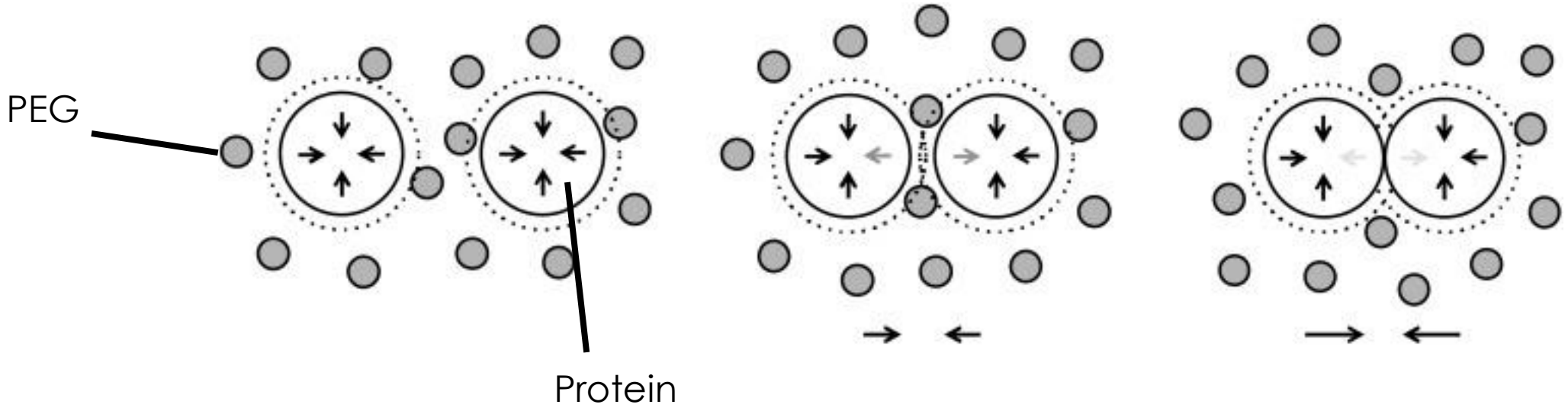
III. Precipitation: c) polymer-induced

- Neutral polymers such as PEG and Dextran can lead to protein precipitation following a principle similar to salting out: polymers at high concentration **attract water molecules away from the solvation shell of proteins** (they act like "**solvent sponges**") leading to aggregation

- **Not to be confused with aqueous two-phase extraction**

III. Precipitation: c) polymer-induced

- In addition, PEG can also lead to protein aggregation of large globular proteins by inducing **depletion forces** first explained by Asakura and Oosawa
- This entropic effect arises because small particles (PEG) are excluded from the volume between two adjacent large particles (proteins)
- There is therefore a larger osmotic pressure from the polymer outside this gap than inside, leading to attractive forces between the proteins



Asakura and Oosawa, On Interaction between Two Bodies Immersed in a Solution of Macromolecules, J. Chem. Phys. 22, 1255 (1954)

III. Precipitation: d) solvent-induced

- The feed is mixed with a **miscible solvent**, e.g. ethanol or methanol
- **Different from solvent extraction, where we used immiscible solvents**
- In sufficiently high molar fraction, the solvent **displaces the water molecules from the solvation shell** around the proteins, disrupting it and casing the proteins to aggregate, similar to what happens with salting out and polymer-induced precipitation
- To be more precise, ethanol or methanol will decrease the dielectric constant of water (but this is beyond the scope of the course)
- Often used for primary isolation of antibodies